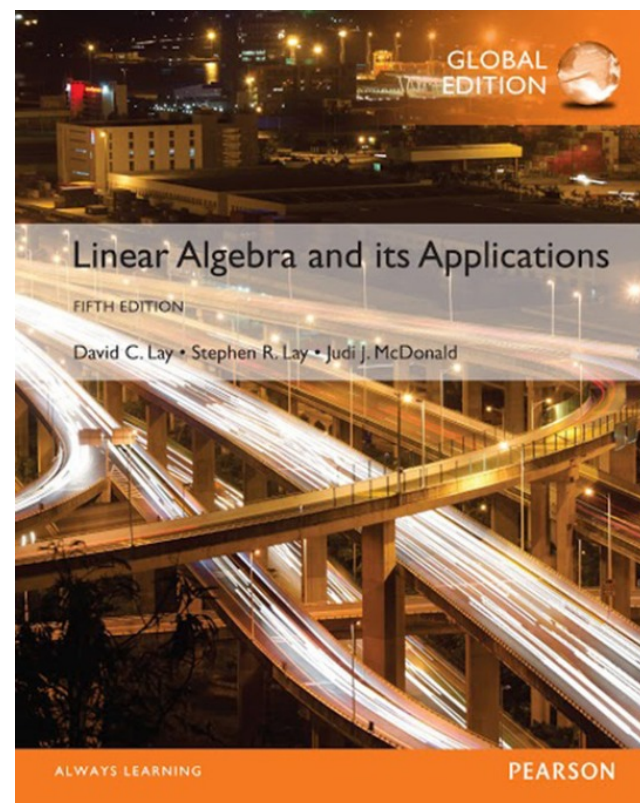


5

Eigenvalues and Eigenvectors

5.1

EIGENVECTORS AND EIGENVALUES



EIGENVECTORS AND EIGENVALUES

- **Definition:** An **eigenvector** of an $n \times n$ matrix A is a nonzero vector $\mathbf{x}$ such that $A\mathbf{x} = \lambda\mathbf{x}$ for some scalar λ . A scalar λ is called an **eigenvalue** of A if there is a nontrivial solution $\mathbf{x}$ of $A\mathbf{x} = \lambda\mathbf{x}$; such an $\mathbf{x}$ is called an *eigenvector corresponding to λ* .
- λ is an eigenvalue of an $n \times n$ matrix A if and only if the equation

$$(A - \lambda I)\mathbf{x} = 0 \quad (3)$$

has a nontrivial solution.

- The set of *all* solutions of (3) is just the null space of the matrix $A - \lambda I$.

EIGENVECTORS AND EIGENVALUES

- So this set is a *subspace* of $\mathbb{R}^n$ and is called the **eigenspace** of A corresponding to λ .
- The eigenspace consists of the zero vector and all the eigenvectors corresponding to λ .
- **Example 3:** Show that 7 is an eigenvalue of matrix

$$A = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix} \text{ and find the corresponding eigenvectors.}$$

EIGENVECTORS AND EIGENVALUES

- **Solution:** The scalar 7 is an eigenvalue of A if and only if the equation

$$A\mathbf{x} = 7\mathbf{x} \quad (1)$$

has a nontrivial solution.

- But (1) is equivalent to $A\mathbf{x} - 7\mathbf{x} = \mathbf{0}$, or

$$(A - 7I)\mathbf{x} = \mathbf{0} \quad (2)$$

- To solve this homogeneous equation, form the matrix

$$A - 7I = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix} - \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix} = \begin{bmatrix} -6 & 6 \\ 5 & -5 \end{bmatrix}$$

EIGENVECTORS AND EIGENVALUES

- The columns of $A - 7I$ are obviously linearly dependent, so (2) has nontrivial solutions.
- To find the corresponding eigenvectors, use row operations:

$$\begin{bmatrix} -6 & 6 & 0 \\ 5 & -5 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

- The general solution has the form $x_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.
- Each vector of this form with $x_2 \neq 0$ is an eigenvector corresponding to $\lambda = 7$.

EIGENVECTORS AND EIGENVALUES

- **Example 4:** Let $A = \begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix}$. An eigenvalue of A is 2. Find a basis for the corresponding eigenspace.

- **Solution:** Form

$$A - 2I = \begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix} - \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 2 & -1 & 6 \\ 2 & -1 & 6 \\ 2 & -1 & 6 \end{bmatrix}$$

and row reduce the augmented matrix for $(A - 2I)\mathbf{x} = \mathbf{0}$.

EIGENVECTORS AND EIGENVALUES

$$\begin{bmatrix} 2 & -1 & 6 & 0 \\ 2 & -1 & 6 & 0 \\ 2 & -1 & 6 & 0 \end{bmatrix} \sim \begin{bmatrix} 2 & -1 & 6 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

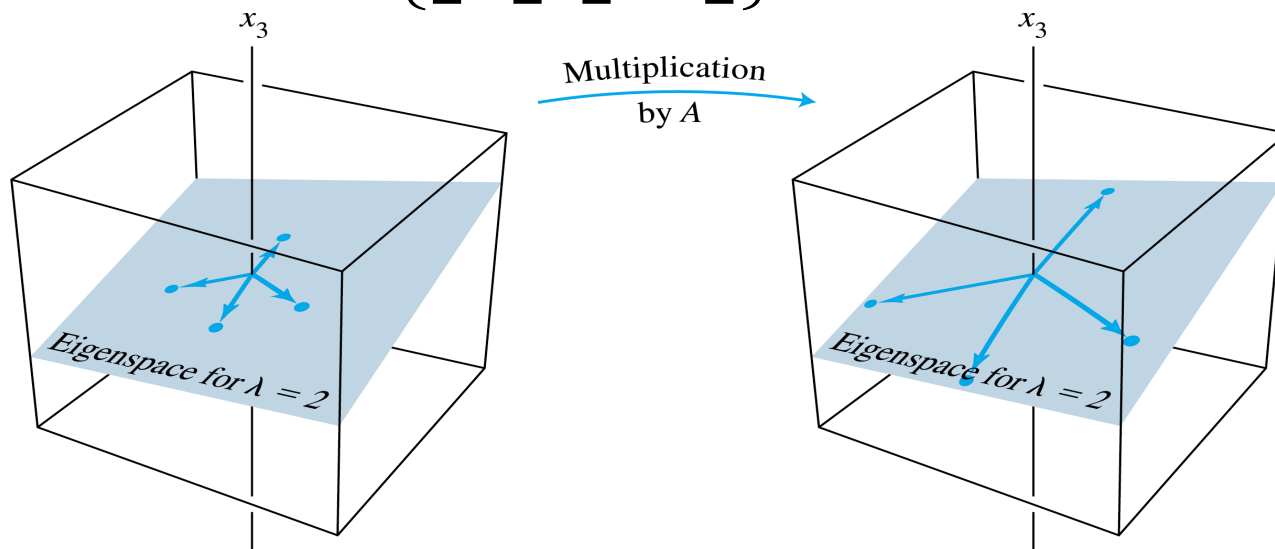
- At this point, it is clear that 2 is indeed an eigenvalue of A because the equation $(A - 2I)\mathbf{x} = 0$ has free variables.
- The general solution is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} 1/2 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}, \text{ } x_2 \text{ and } x_3 \text{ free.}$$

EIGENVECTORS AND EIGENVALUES

- The eigenspace, shown in the following figure, is a two-dimensional subspace of $\mathbb{R}^3$. A basis is

$$\left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} \right\}$$



A acts as a dilation on the eigenspace.

EIGENVECTORS AND EIGENVALUES

- **Theorem 1:** The eigenvalues of a triangular matrix are the entries on its main diagonal.
- **Proof:** For simplicity, consider the 3×3 case.
- If A is upper triangular, the $A - \lambda I$ has the form

$$\begin{aligned} A - \lambda I &= \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} \\ &= \begin{bmatrix} a_{11} - \lambda & a_{12} & a_{13} \\ 0 & a_{22} - \lambda & a_{23} \\ 0 & 0 & a_{33} - \lambda \end{bmatrix} \end{aligned}$$

EIGENVECTORS AND EIGENVALUES

- The scalar λ is an eigenvalue of A if and only if the equation $(A - \lambda I)\mathbf{x} = \mathbf{0}$ has a nontrivial solution, that is, if and only if the equation has a free variable.
- Because of the zero entries in $A - \lambda I$, it is easy to see that $(A - \lambda I)\mathbf{x} = \mathbf{0}$ has a free variable if and only if at least one of the entries on the diagonal of $A - \lambda I$ is zero.
- This happens if and only if λ equals one of the entries a_{11} , a_{22} , a_{33} in A .

EIGENVECTORS AND EIGENVALUES

- **Theorem 2:** If $\mathbf{v}_1, \dots, \mathbf{v}_r$ are eigenvectors that correspond to distinct eigenvalues $\lambda_1, \dots, \lambda_r$ of an $n \times n$ matrix A , then the set $\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$ is linearly independent.
- **Proof:** Suppose $\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$ is linearly dependent.
- Since $\mathbf{v}_1$ is nonzero, Theorem 7 in Section 1.7 says that one of the vectors in the set is a linear combination of the preceding vectors.
- Let p be the least index such that $\mathbf{v}_{p+1}$ is a linear combination of the preceding (linearly independent) vectors.

EIGENVECTORS AND EIGENVALUES

- Then there exist scalars $c_1, \dots, c_p$ such that

$$(5) \quad c_1 \mathbf{v}_1 + \cdots + c_p \mathbf{v}_p = \mathbf{v}_{p+1}$$

- Multiplying both sides of (5) by A and using the fact that $A\mathbf{v}_k = \lambda_k \mathbf{v}_k$ for each k , we obtain

$$c_1 A\mathbf{v}_1 + \cdots + c_p A\mathbf{v}_p = A\mathbf{v}_{p+1}$$

$$c_1 \lambda_1 \mathbf{v}_1 + \cdots + c_p \lambda_p \mathbf{v}_p = \lambda_{p+1} \mathbf{v}_{p+1} \quad (6)$$

- Multiplying both sides of (5) by λ_{p+1} and subtracting the result from (6), we have

$$c_1 (\lambda_1 - \lambda_{p+1}) \mathbf{v}_1 + \cdots + c_p (\lambda_p - \lambda_{p+1}) \mathbf{v}_p = 0 \quad (7)$$

EIGENVECTORS AND EIGENVALUES

- Since $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is linearly independent, the weights in (7) are all zero.
- But none of the factors $\lambda_i - \lambda_{p+1}$ are zero, because the eigenvalues are distinct.
- Hence $c_i = 0$ for $i = 1, \dots, p$.
- But then (5) says that $\mathbf{v}_{p+1} = 0$, which is impossible.

EIGENVECTORS AND DIFFERENCE EQUATIONS

- Hence $\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$ cannot be linearly dependent and therefore must be linearly independent.
- If A is an $n \times n$ matrix, then (8) is a *recursive* description of a sequence $\{x_k\}$ in $\mathbb{R}^n$.

$$\mathbf{x}_{k+1} = A\mathbf{x}_k \quad (k = 0, 1, 2, \dots) \quad (8)$$

- A **solution** of (8) is an explicit description of $\{x_k\}$ whose formula for each x_k does not depend directly on A or on the preceding terms in the sequence other than the initial term $\mathbf{x}_0$.

EIGENVECTORS AND DIFFERENCE EQUATIONS

- The simplest way to build a solution of (8) is to take an eigenvector $\mathbf{x}_0$ and its corresponding eigenvalue λ and let

$$\mathbf{x}_k = \lambda^k \mathbf{x}_0 \quad (k = 1, 2, \dots) \quad (9)$$

- This sequence is a solution because

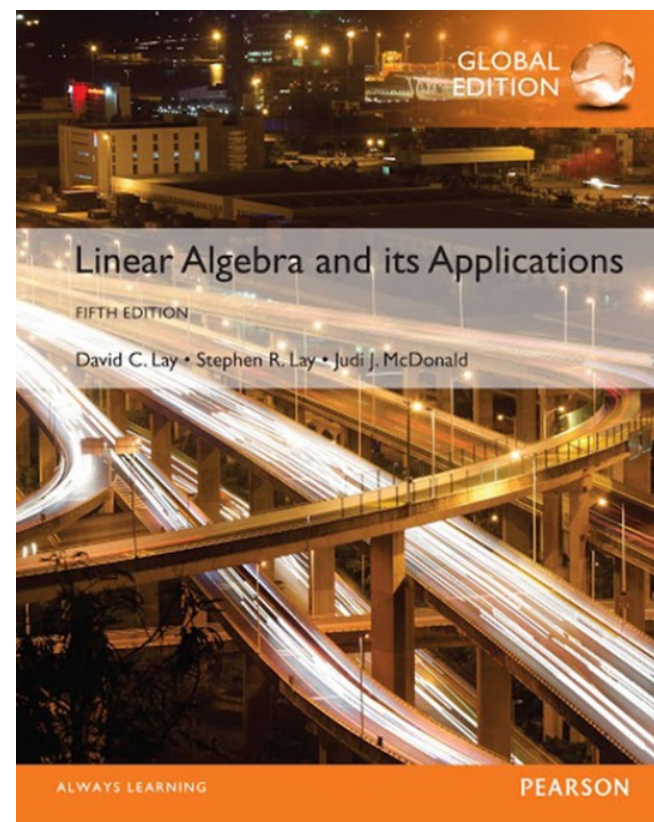
$$A\mathbf{x}_k = A(\lambda^k \mathbf{x}_0) = \lambda^k (A\mathbf{x}_0) = \lambda^k (\lambda \mathbf{x}_0) = \lambda^{k+1} \mathbf{x}_0 = \mathbf{x}_{k+1}$$

5

Eigenvalues and Eigenvectors

5.2

THE CHARACTERISTIC EQUATION



DETERMINANTS

- Let A be an $n \times n$ matrix, let U be any echelon form obtained from A by row replacements and row interchanges (without scaling), and let r be the number of such row interchanges.
- Then the **determinant** of A , written as $\det A$, is $(-1)^r$ times the product of the diagonal entries $u_{11}, \dots, u_{nn}$ in U .
- If A is invertible, then $u_{11}, \dots, u_{nn}$ are all pivots (because $A \sim I_n$ and the u_{ii} have not been scaled to 1's).

DETERMINANTS

- Otherwise, at least u_{nn} is zero, and the product $u_{11} \dots u_{nn}$ is zero.

- Thus

$$\det A = \begin{cases} (-1)^r \cdot \left(\text{product of pivots in } U \right), & \text{when } A \text{ is invertible} \\ 0, & \text{when } A \text{ is not invertible} \end{cases}$$

DETERMINANTS

- **Example 1:** Compute $\det A$ for $A = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$.

- **Solution:** The following row reduction uses one row interchange:

$$A \sim \begin{bmatrix} 1 & 5 & 0 \\ 0 & -6 & -1 \\ 0 & -2 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 5 & 0 \\ 0 & -2 & 0 \\ 0 & -6 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 5 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -1 \end{bmatrix} = U_1$$

DETERMINANTS

- So $\det A$ equals $(-1)^1(1)(-2)(-1) = -2$.
- The following alternative row reduction avoids the row interchange and produces a different echelon form.

- The last step adds $-1/3$ times row 2 to row 3:

$$A \sim \begin{bmatrix} 1 & 5 & 0 \\ 0 & -6 & -1 \\ 0 & -2 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 5 & 0 \\ 0 & -6 & -1 \\ 0 & 0 & 1/3 \end{bmatrix} = U_2$$

- This time $\det A$ is $(-1)^0(1)(-6)(1/3) = -2$, the same as before.

THE INVERTIBLE MATRIX THEOREM (CONTINUED)

- **Theorem:** Let A be an $n \times n$ matrix. Then A is invertible if and only if:
 - s. The number 0 is *not* an eigenvalue of A .
 - t. The determinant of A is *not* zero.
- **Theorem 3: Properties of Determinants**
- Let A and B be $n \times n$ matrices.
 - a. A is invertible if and only if $\det A \neq 0$
 - b. $\det AB = (\det A)(\det B)$.
 - c. $\det A^T = \det A$.

PROPERTIES OF DETERMINANTS

- d. If A is triangular, then $\det A$ is the product of the entries on the main diagonal of A .
- e. A row replacement operation on A does not change the determinant. A row interchange changes the sign of the determinant. A row scaling also scales the determinant by the same scalar factor.

THE CHARACTERISTIC EQUATION

- Theorem 3(a) shows how to determine when a matrix of the form $A - \lambda I$ is *not* invertible.
- The scalar equation $\det(A - \lambda I) = 0$ is called the **characteristic equation** of A .
- A scalar λ is an eigenvalue of an $n \times n$ matrix A if and only if λ satisfies the characteristic equation

$$\det(A - \lambda I) = 0$$

THE CHARACTERISTIC EQUATION

- **Example 3:** Find the characteristic equation of

$$A = \begin{bmatrix} 5 & -2 & 6 & -1 \\ 0 & 3 & -8 & 0 \\ 0 & 0 & 5 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- **Solution:** Form $A - \lambda I$, and use Theorem 3(d):

THE CHARACTERISTIC EQUATION

$$\det(A - \lambda I) = \det \begin{bmatrix} 5 - \lambda & -2 & 6 & -1 \\ 0 & 3 - \lambda & -8 & 0 \\ 0 & 0 & 5 - \lambda & 4 \\ 0 & 0 & 0 & 1 - \lambda \end{bmatrix}$$
$$= (5 - \lambda)(3 - \lambda)(5 - \lambda)(1 - \lambda)$$

- The characteristic equation is

$$(5 - \lambda)^2 (3 - \lambda)(1 - \lambda) = 0$$

or

$$(\lambda - 5)^2 (\lambda - 3)(\lambda - 1) = 0$$

THE CHARACTERISTIC EQUATION

- Expanding the product, we can also write

$$\lambda^4 - 14\lambda^3 + 68\lambda^2 - 130\lambda + 75 = 0$$

- If A is an $n \times n$ matrix, then $\det(A - \lambda I)$ is a polynomial of degree n called the **characteristic polynomial** of A .
- The eigenvalue 5 in Example 3 is said to have *multiplicity* 2 because $(\lambda - 5)$ occurs two times as a factor of the characteristic polynomial.
- In general, the **(algebraic) multiplicity** of an eigenvalue λ is its multiplicity as a root of the characteristic equation.

SIMILARITY

- If A and B are $n \times n$ matrices, then A is **similar to** B if there is an invertible matrix P such that $P^{-1}AP = B$, or, equivalently, $A = PBP^{-1}$.
- Writing Q for P^{-1} , we have $Q^{-1}BQ = A$.
- So B is also similar to A , and we say simply that A and B **are similar**.
- Changing A into $P^{-1}AP$ is called a **similarity transformation**.

SIMILARITY

- **Theorem 4:** If $n \times n$ matrices A and B are similar, then they have the same characteristic polynomial and hence the same eigenvalues (with the same multiplicities).

- **Proof:** If $B = P^{-1}AP$ then,

$$B - \lambda I = P^{-1}AP - \lambda P^{-1}P = P^{-1}(AP - \lambda P) = P^{-1}(A - \lambda I)P$$

- Using the multiplicative property (b) in Theorem (3), we compute

$$\begin{aligned}\det(B - \lambda I) &= \det[P^{-1}(A - \lambda I)P] \\ &= \det(P^{-1}) \cdot \det(A - \lambda I) \cdot \det(P) \quad (2)\end{aligned}$$

SIMILARITY

- Since $\det(P^{-1}) \cdot \det(P) = \det(P^{-1}P) = \det I = 1$, we see from equation (1) that $\det(B - \lambda I) = \det(A - \lambda I)$.

- **Warnings:**

1. The matrices

$$\begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} \text{ and } \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

are not similar even though they have the same eigenvalues.

SIMILARITY

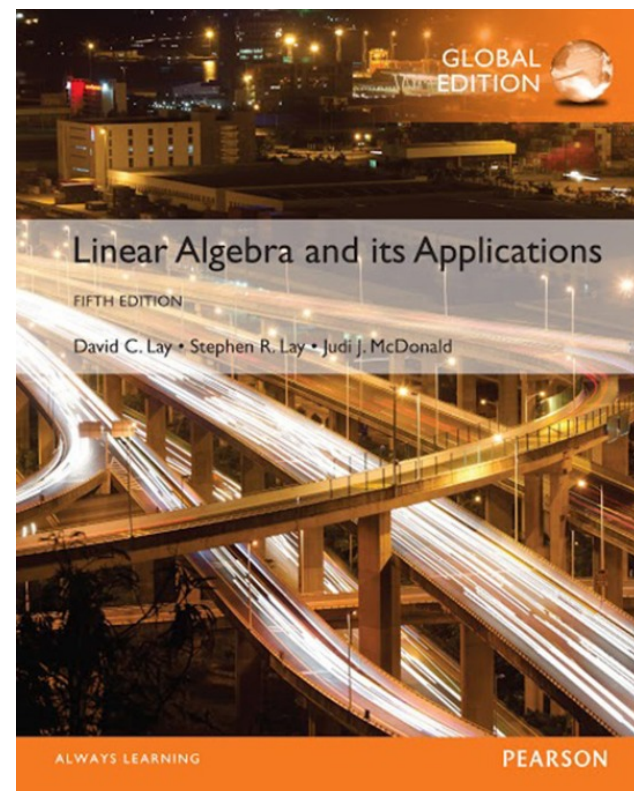
2. Similarity is not the same as row equivalence. (If A is row equivalent to B , then $B = EA$ for some invertible matrix E). Row operations on a matrix usually change its eigenvalues.

5

Eigenvalues and Eigenvectors

5.3

DIAGONALIZATION



DIAGONALIZATION

- **Example 2:** Let $A = \begin{bmatrix} 7 & 2 \\ -4 & 1 \end{bmatrix}$. Find a formula for

A^k , given that $A = PDP^{-1}$, where

$$P = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix} \quad \text{and} \quad D = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}$$

- **Solution:** The standard formula for the inverse of a 2×2 matrix yields

$$P^{-1} = \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix}$$

DIAGONALIZATION

- Then, by associativity of matrix multiplication,

$$A^2 = (PDP^{-1})(PDP^{-1}) = PD \underbrace{(P^{-1}P)}_I DP^{-1} = PDDP^{-1}$$

$$= PD^2P^{-1} = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} 5^2 & 0 \\ 0 & 3^2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix}$$

- Again,

$$A^3 = (PDP^{-1})A^2 = (PD \underbrace{P^{-1}P}_I) D^2 P^{-1} = PDD^2P^{-1} = PD^3P^{-1}$$

DIAGONALIZATION

- In general, for $k \geq 1$,

$$\begin{aligned} A^k &= PD^k P^{-1} = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} 5^k & 0 \\ 0 & 3^k \end{bmatrix} \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 2 \cdot 5^k - 3^k & 5^k - 3^k \\ 2 \cdot 3^k - 2 \cdot 5^k & 2 \cdot 3^k - 5^k \end{bmatrix} \end{aligned}$$

- A square matrix A is said to be **diagonalizable** if A is similar to a diagonal matrix, that is, if $A = PDP^{-1}$ for some invertible matrix P and some diagonal, matrix D .

THE DIAGONALIZATION THEOREM

- **Theorem 5:** An $n \times n$ matrix A is diagonalizable if and only if A has n linearly independent eigenvectors.

In fact, $A = PDP^{-1}$, with D a diagonal matrix, if and only if the columns of P are n linearly independent eigenvectors of A . In this case, the diagonal entries of D are eigenvalues of A that correspond, respectively, to the eigenvectors in P .

In other words, A is diagonalizable if and only if there are enough eigenvectors to form a basis of $\mathbb{R}^n$. We call such a basis an **eigenvector basis** of $\mathbb{R}^n$.

THE DIAGONALIZATION THEOREM

- **Proof:** First, observe that if P is any $n \times n$ matrix with columns $\mathbf{v}_1, \dots, \mathbf{v}_n$, and if D is any diagonal matrix with diagonal entries $\lambda_1, \dots, \lambda_n$, then

$$AP = A \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \end{bmatrix} = \begin{bmatrix} A\mathbf{v}_1 & A\mathbf{v}_2 & \cdots & A\mathbf{v}_n \end{bmatrix} \quad (1)$$

while

$$PD = P \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix} = \begin{bmatrix} \lambda_1 \mathbf{v}_1 & \lambda_2 \mathbf{v}_2 & \cdots & \lambda_n \mathbf{v}_n \end{bmatrix} \quad (2)$$

THE DIAGONALIZATION THEOREM

- Now suppose A is diagonalizable and $A = PDP^{-1}$. Then right-multiplying this relation by P , we have
 $AP = PD$.
- In this case, equations (1) and (2) imply that
$$\begin{bmatrix} A\mathbf{v}_1 & A\mathbf{v}_2 & \cdots & A\mathbf{v}_n \end{bmatrix} = \begin{bmatrix} \lambda_1 \mathbf{v}_1 & \lambda_2 \mathbf{v}_2 & \cdots & \lambda_n \mathbf{v}_n \end{bmatrix} \quad (3)$$
- Equating columns, we find that
$$A\mathbf{v}_1 = \lambda_1 \mathbf{v}_1, A\mathbf{v}_2 = \lambda_2 \mathbf{v}_2, \dots, A\mathbf{v}_n = \lambda_n \mathbf{v}_n \quad (4)$$
- Since P is invertible, its columns $\mathbf{v}_1, \dots, \mathbf{v}_n$ must be linearly independent.

THE DIAGONALIZATION THEOREM

- Also, since these columns are nonzero, the equations in (4) show that $\lambda_1, \dots, \lambda_n$ are eigenvalues and $\mathbf{v}_1, \dots, \mathbf{v}_n$ are corresponding eigenvectors.
- This argument proves the “only if” parts of the first and second statements, along with the third statement, of the theorem.
- Finally, given any n eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_n$, use them to construct the columns of P and use corresponding eigenvalues $\lambda_1, \dots, \lambda_n$ to construct D .

THE DIAGONALIZATION THEOREM

- By equations (1)–(3), $AP = PD$.
- This is true without any condition on the eigenvectors.
- If, in fact, the eigenvectors are linearly independent, then P is invertible (by the Invertible Matrix Theorem), and $AP = PD$ implies that $A = PDP^{-1}$.

DIAGONALIZING MATRICES

- **Example 3:** Diagonalize the following matrix, if possible.

$$A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}$$

That is, find an invertible matrix P and a diagonal matrix D such that $A = PDP^{-1}$.

- **Solution:** There are four steps to implement the description in Theorem 5.
- *Step 1. Find the eigenvalues of A .*
- Here, the characteristic equation turns out to involve a cubic polynomial that can be factored:

DIAGONALIZING MATRICES

$$\begin{aligned} 0 &= \det(A - \lambda I) = -\lambda^3 - 3\lambda^2 + 4 \\ &= -(\lambda - 1)(\lambda + 2)^2 \end{aligned}$$

- The eigenvalues are $\lambda = 1$ and $\lambda = -2$.
- *Step 2. Find three linearly independent eigenvectors of A .*
- *Three* vectors are needed because A is a 3×3 matrix.
- This is a critical step.
- If it fails, then Theorem 5 says that A cannot be diagonalized.

DIAGONALIZING MATRICES

- Basis for $\lambda = 1$: $\mathbf{v}_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$
- Basis for $\lambda = -2$: $\mathbf{v}_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$ and $\mathbf{v}_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$
- You can check that $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is a linearly independent set.

DIAGONALIZING MATRICES

- *Step 3. Construct P from the vectors in step 2.*
- The order of the vectors is unimportant.
- Using the order chosen in step 2, form

$$P = \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 \end{bmatrix} = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

- *Step 4. Construct D from the corresponding eigenvalues.*
- In this step, it is essential that the order of the eigenvalues matches the order chosen for the columns of P .

DIAGONALIZING MATRICES

- Use the eigenvalue $\lambda = -2$ twice, once for each of the eigenvectors corresponding to $\lambda = -2$:

$$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

- To avoid computing P^{-1} , simply verify that $AP = PD$.
- Compute

$$AP = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 2 \\ -1 & -2 & 0 \\ 1 & 0 & -2 \end{bmatrix}$$

DIAGONALIZING MATRICES

$$PD = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 2 \\ -1 & -2 & 0 \\ 1 & 0 & -2 \end{bmatrix}$$

- **Theorem 6:** An $n \times n$ matrix with n distinct eigenvalues is diagonalizable.
- **Proof:** Let $\mathbf{v}_1, \dots, \mathbf{v}_n$ be eigenvectors corresponding to the n distinct eigenvalues of a matrix A .
- Then $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is linearly independent, by Theorem 2 in Section 5.1.
- Hence A is diagonalizable, by Theorem 5.

MATRICES WHOSE EIGENVALUES ARE NOT DISTINCT

- It is not *necessary* for an $n \times n$ matrix to have n distinct eigenvalues in order to be diagonalizable.
- The 3×3 matrix in Example 3 is diagonalizable even though it has only two distinct eigenvalues.
- If an $n \times n$ matrix A has n distinct eigenvalues, with corresponding eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_n$, and if $P = [\mathbf{v}_1 \quad \cdots \quad \mathbf{v}_n]$, then P is automatically invertible because its columns are linearly independent, by Theorem 2.

MATRICES WHOSE EIGENVALUES ARE NOT DISTINCT

- When A is diagonalizable but has fewer than n distinct eigenvalues, it is still possible to build P in a way that makes P automatically invertible, as the next theorem shows.
- **Theorem 7:** Let A be an $n \times n$ matrix whose distinct eigenvalues are $\lambda_1, \dots, \lambda_p$.
 - a. For $1 \leq k \leq p$, the dimension of the eigenspace for λ_k is less than or equal to the multiplicity of the eigenvalue λ_k .

MATRICES WHOSE EIGENVALUES ARE NOT DISTINCT

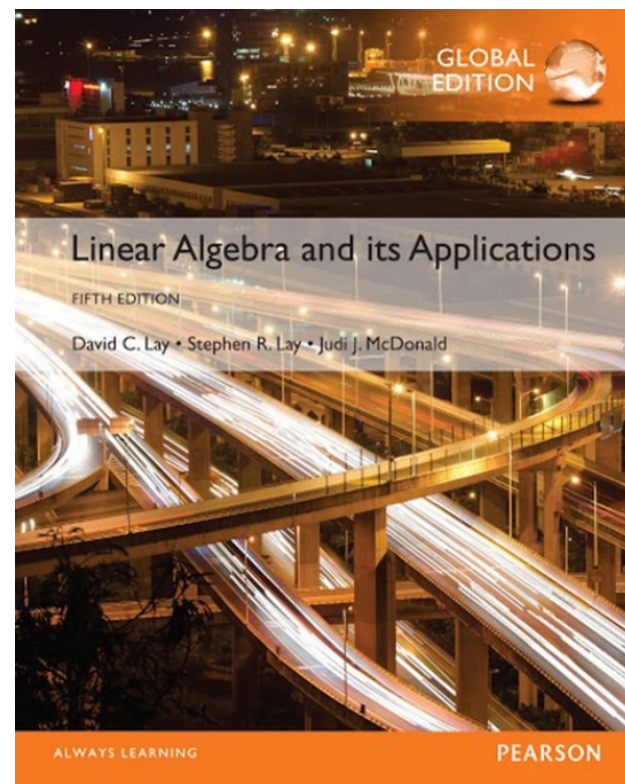
- b. The matrix A is diagonalizable if and only if the sum of the dimensions of the eigenspaces equals n , and this happens if and only if (i) the characteristic polynomial factors completely into linear factors and (ii) the dimension of the eigenspace for each λ_k equals the multiplicity of λ_k .
- c. If A is diagonalizable and B_k is a basis for the eigenspace corresponding to λ_k for each k , then the total collection of vectors in the sets $B_1, \dots, B_p$ forms an eigenvector basis for $\mathbb{R}^n$.

5

Eigenvalues and Eigenvectors

5.4

EIGENVECTORS AND LINEAR TRANSFORMATIONS



THE MATRIX OF A LINEAR TRANSFORMATION

- Given any x in V , the coordinate vector $[x]_{\beta}$ is in $\mathbb{R}^n$ and the coordinate vector of its image, $[T(x)]_{\mathcal{C}}$ is in $\mathbb{R}^m$, as shown in Fig. 1 below.

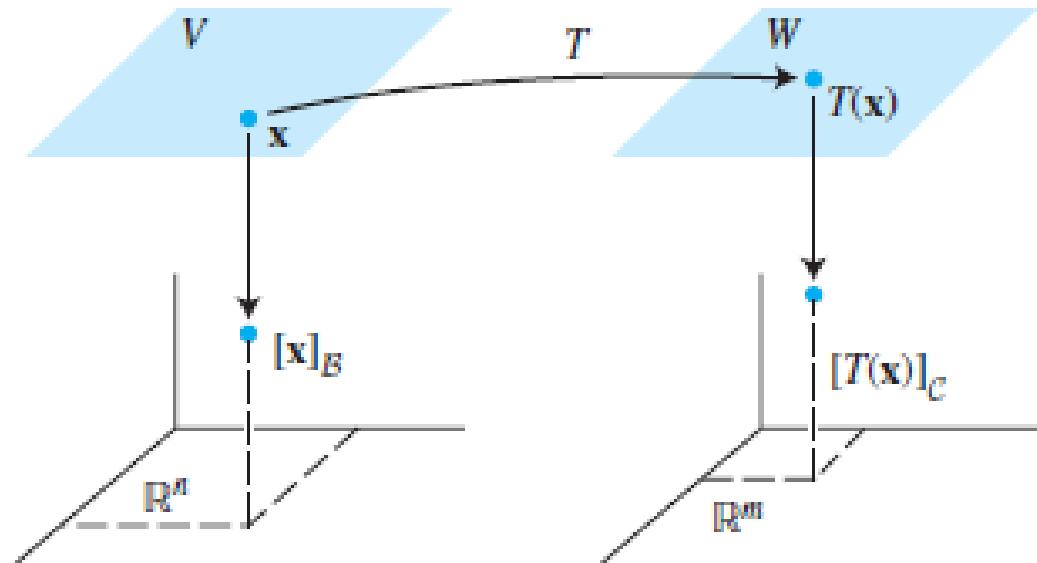


FIGURE 1 A linear transformation from V to W .

THE MATRIX OF A LINEAR TRANSFORMATION

- The connection between $[x]_{\beta}$ and $[T(x)]_{\mathcal{C}}$ is easy to find. Let $\{b_1, \dots, b_n\}$ be the basis β for V . If $x = r_1b_1 + \dots + r_nb_n$, then,

$$[x]_{\beta} = \begin{bmatrix} r_1 \\ \vdots \\ r_n \end{bmatrix}$$

- And

$$T(x) = T(r_1b_1 + \dots + r_nb_n) = r_1T(b_1) + \dots + r_nT(b_n) \quad (1)$$

- because T is linear.

THE MATRIX OF A LINEAR TRANSFORMATION

- Now, since the coordinate mapping from W to $\mathbb{R}^m$ is linear, equation (1) leads to

$$[T(x)]_C = r_1[T(b_1)]_C + \cdots + r_n[T(b_n)]_C \quad (2)$$

- Since C -coordinate vectors are in $\mathbb{R}^m$, the vector equation (2) can be written as a matrix equation, namely,

$$[T(x)]_C = M[x]_\beta \quad (3)$$

- where

$$M = [[T(b_1)]_C \ [T(b_2)]_C \ \cdots \ [T(b_n)]_C] \quad (4)$$

THE MATRIX OF A LINEAR TRANSFORMATION

- The matrix M is a matrix representation of T , called the matrix for T relative to the bases β and C . See Fig. 2 below:

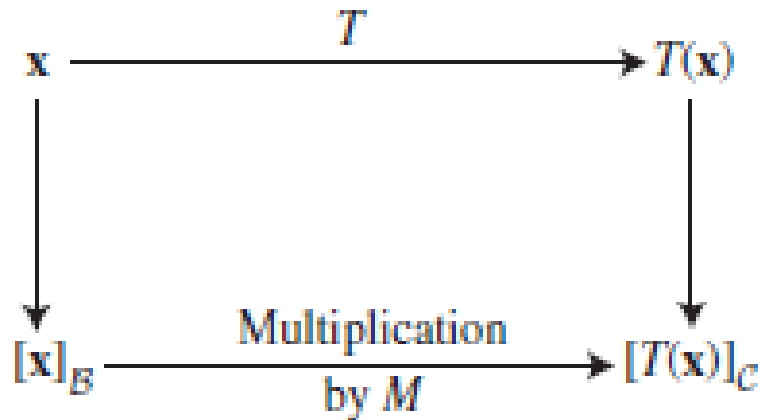


FIGURE 2

THE MATRIX OF A LINEAR TRANSFORMATION

- **Example 1** Suppose $\beta = \{b_1, b_2\}$ is a basis for V and $C = \{c_1, c_2, c_3\}$ is a basis for W . Let $T: V \rightarrow W$ be a linear transformation with the property that

$$T(b_1) = 3c_1 - 2c_2 + 5c_3 \quad \text{and} \quad T(b_2) = 4c_1 + 7c_2 - c_3$$

- Find the matrix M for T relative to β and C .

THE MATRIX OF A LINEAR TRANSFORMATION

- **Solution** The C -coordinate vectors of the images of b_1 and b_2 are

$$[T(b_1)]_C = \begin{bmatrix} 3 \\ -2 \\ 5 \end{bmatrix} \text{ and } [T(b_2)]_C = \begin{bmatrix} 4 \\ 7 \\ -1 \end{bmatrix}$$

- Hence

$$M = \begin{bmatrix} 3 & 4 \\ -2 & 7 \\ 5 & -1 \end{bmatrix}$$

- If β and C are bases for the same space V and if T is the identity transformation $T(x) = x$ for x in V , then matrix M in (4) is just a change-of-coordinates matrix.

LINEAR TRANSFORMATIONS FROM V INTO V

- In the common case where W is the same V and the basis C is the same as β , then the matrix M in (4) is called the **matrix for T relative to β** , or simply the **β -matrix for T** , and is denoted by $[T]_{\beta}$.
- See Fig. 3 below

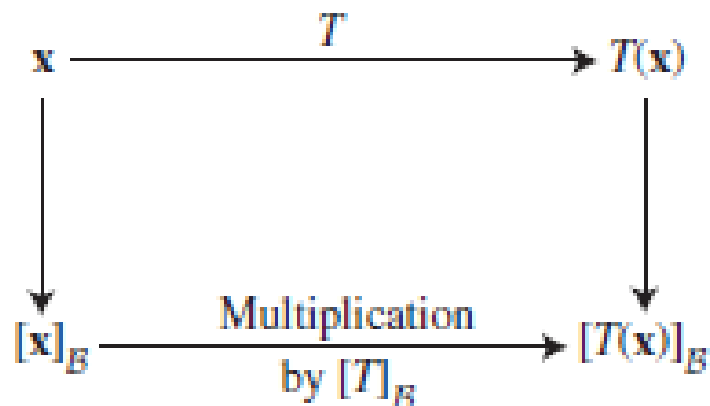


FIGURE 3

LINEAR TRANSFORMATIONS FROM V INTO V

- **Example 2** The mapping $T : \mathbb{P}_2 \rightarrow \mathbb{P}_2$ defined by

$$T(a_0 + a_1t + a_2t^2) = a_1 + 2a_2t$$

- is a linear transformation.
- a) Find the β -matrix for T , when β is the basis $\{1, t, t^2\}$
 - b) Verify that $[T(p)]_\beta = [T]_\beta [p]_\beta$

LINEAR TRANSFORMATIONS FROM V INTO V

- **Solution**

a) Compute the images of the basis vectors:

$$T(1) = 0$$

$$T(t) = 1$$

$$T(t^2) = 2t$$

- Then write the β -coordinate vectors of $T(1)$, $T(t)$, and $T(t^2)$ and place them together as the β -matrix for T :

LINEAR TRANSFORMATIONS FROM V INTO V

$$[T(1)]_{\beta} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \quad [T(t)]_{\beta} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad [T(t^2)]_{\beta} = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}$$

$$[T]_{\beta} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

b) For a general $p(t) = a_0 + a_1 t + a_2 t^2$,

$$[T(p)]_{\beta} = [a_1 + 2a_2 t]_{\beta} = \begin{bmatrix} a_1 \\ 2a_2 \\ 0 \end{bmatrix}$$

LINEAR TRANSFORMATIONS FROM V INTO V

$$= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = [T]_{\beta} [p]_{\beta}$$

- See Fig. 4 below:

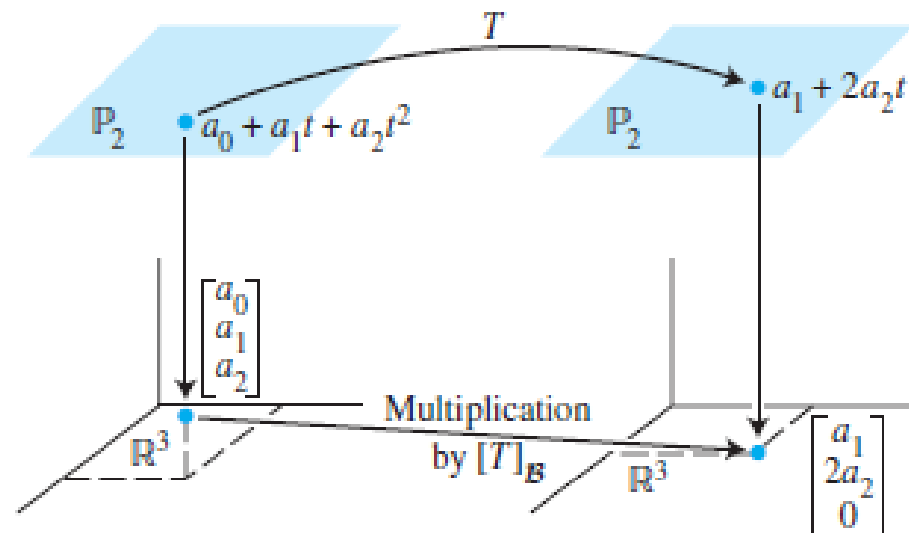


FIGURE 4 Matrix representation of a linear transformation.

LINEAR TRANSFORMATIONS ON $\mathbb{R}^n$

- **Theorem 8:** Suppose $A = PDP^{-1}$, where D is a diagonal $n \times n$ matrix. If β is the basis for $\mathbb{R}^n$ formed from the columns of P , then D is the β -matrix for the transformation $x \mapsto Ax$.
- **Proof** Denote the columns of P by $b_1, \dots, b_n$, so that $\beta = \{b_1, \dots, b_n\}$ and $P = [b_1 \dots b_n]$. In this case, P is the change-of-coordinates matrix P_β discussed in Section 4.4, where

$$P[x]_\beta = x \quad \text{and} \quad [x]_\beta = P^{-1}x$$

LINEAR TRANSFORMATIONS ON $\mathbb{R}^n$

- If $T(x) = Ax$ for x in $\mathbb{R}^n$, then

$$\begin{aligned}[T]_{\beta} &= [T(b_1)]_{\beta} \quad \dots \quad [T(b_n)]_{\beta} && \text{Definition of } [T]_{\beta} \\ &= [Ab_1]_{\beta} \quad \dots \quad [Ab_n]_{\beta} && \text{Since } T(x) = Ax \\ &= [P^{-1}Ab_1 \quad \dots \quad P^{-1}Ab_n] && \text{Change of coordinates} \\ &= P^{-1}A[b_1 \quad \dots \quad b_n] && \text{Matrix multiplication} \\ &= P^{-1}AP\end{aligned}$$

- Since $A = PDP^{-1}$, we have $[T]_{\beta} = P^{-1}AP = D$.

LINEAR TRANSFORMATIONS ON $\mathbb{R}^n$

- **Example 3** Define $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by $T(\mathbf{x}) = A\mathbf{x}$, where $A = \begin{bmatrix} 7 & 2 \\ -4 & 1 \end{bmatrix}$. Find a basis β for $\mathbb{R}^2$ with the property that the β -matrix for T is a diagonal matrix.
- **Solution** From Example 2 in Section 5.3 we know that $A = PDP^{-1}$, where

$$P = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix} \text{ and } D = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}$$

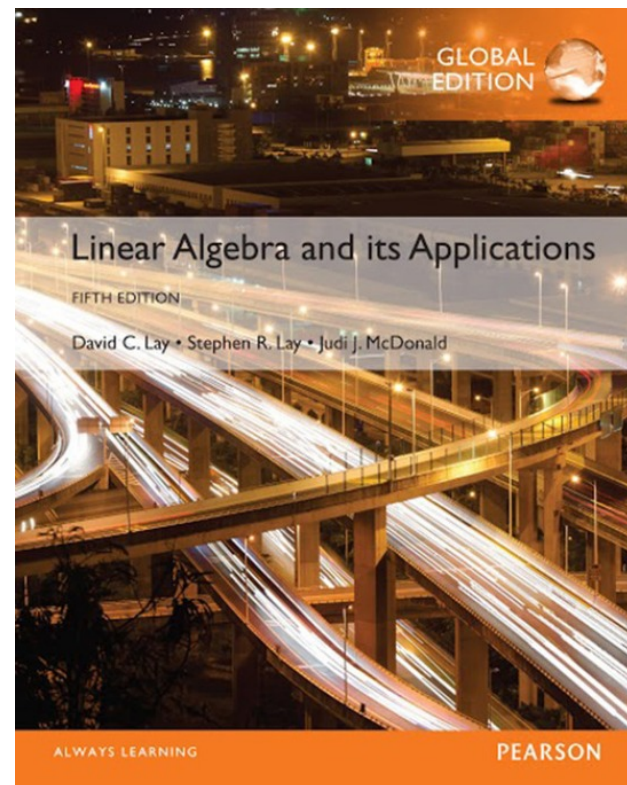
- The columns of P , call them $\mathbf{b}_1$ and $\mathbf{b}_2$, are eigenvectors of A . By Theorem 8, D is the β -matrix for T when $\beta = \{\mathbf{b}_1, \mathbf{b}_2\}$. The mappings $\mathbf{x} \mapsto A\mathbf{x}$ and $\mathbf{u} \mapsto D\mathbf{u}$ describe the same linear transformation, relative to different bases.

5

Eigenvalues and Eigenvectors

5.5

COMPLEX EIGENVALUES



COMPLEX EIGENVALUES

- The matrix eigenvalue-eigenvector theory already developed for $\mathbb{R}^n$ applies equally well to $\mathbb{C}^n$.
- So a complex scalar λ satisfied $\det(A - \lambda I) = 0$ if and only if there is a nonzero vector x in $\mathbb{C}^n$ such that $Ax = \lambda x$.
- We call λ a **(complex) eigenvalue** and x a **(complex) eigenvector** corresponding to λ .

COMPLEX EIGENVALUES

- **Example 1** If $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, then the linear transformation $x \mapsto Ax$ on $\mathbb{R}^2$ rotates the plane counterclockwise through a quarter-turn.
- The action of A is periodic, since after four quarter-turns, a vector is back where it started.
- Obviously, no nonzero vector is mapped into a multiple of itself, so A has no eigenvectors in $\mathbb{R}^2$ and hence no real eigenvalues.
- In fact, the characteristic equation of A is

$$\lambda^2 + 1 = 0$$

COMPLEX EIGENVALUES

- The only roots are complex: $\lambda = i$ and $\lambda = -i$. However, if we permit A to act on $\mathbb{C}^2$, then

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -i \end{bmatrix} = \begin{bmatrix} i \\ 1 \end{bmatrix} = -i \begin{bmatrix} 1 \\ i \end{bmatrix}$$
$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ i \end{bmatrix} = \begin{bmatrix} -i \\ 1 \end{bmatrix} = -i \begin{bmatrix} 1 \\ i \end{bmatrix}$$

- Thus i and $-i$ are eigenvalues, with $\begin{bmatrix} 1 \\ -i \end{bmatrix}$ and $\begin{bmatrix} 1 \\ i \end{bmatrix}$ as corresponding eigenvectors.

REAL AND IMAGINARY PARTS OF VECTORS

- The complex conjugate of a complex vector x in $\mathbb{C}^n$ is the vector $\bar{x}$ in $\mathbb{C}^n$ whose entries are the complex conjugates of the entries in x .
- The **real** and **imaginary parts** of a complex vector x are the vectors $\text{Re } x$ and $\text{Im } x$ in $\mathbb{R}^n$ formed from the real and imaginary parts of the entries of x .

REAL AND IMAGINARY PARTS OF VECTORS

■ **Example 4** If $x = \begin{bmatrix} 3 - i \\ i \\ 2 + 5i \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 2 \end{bmatrix} + i \begin{bmatrix} -1 \\ 1 \\ 5 \end{bmatrix}$, then

$$\operatorname{Re} x = \begin{bmatrix} 3 \\ 0 \\ 2 \end{bmatrix}, \quad \operatorname{Im} x = \begin{bmatrix} -1 \\ 1 \\ 5 \end{bmatrix}, \quad \text{and} \quad \bar{x} = \begin{bmatrix} 3 \\ 0 \\ 2 \end{bmatrix} - i \begin{bmatrix} -1 \\ 1 \\ 5 \end{bmatrix} = \begin{bmatrix} 3 + i \\ -i \\ 2 - 5i \end{bmatrix}$$

EIGENVALUES AND EIGENVECTORS OF A REAL MATRIX THAT ACTS ON $\mathbb{C}^n$

- **Theorem 9:** Let A be a real 2×2 matrix with a complex eigenvalue $\lambda = a - bi$ ($b \neq 0$) and an associated eigenvector v in $\mathbb{C}^2$. Then

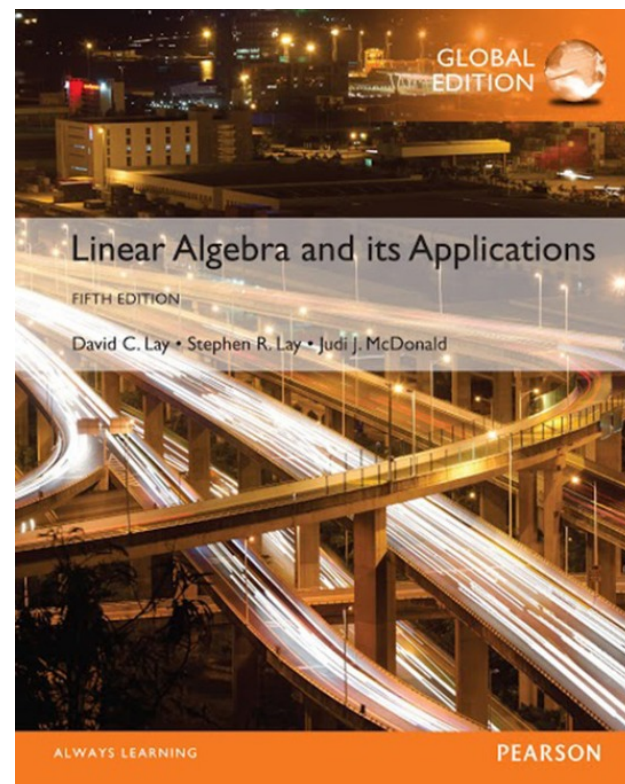
$$A = PCP^{-1}, \text{ where } P = [\operatorname{Re} v \quad \operatorname{Im} v] \text{ and } C = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$$

5

Eigenvalues and Eigenvectors

5.6

DISCRETE DYNAMICAL SYSTEMS



GRAPHICAL DESCRIPTION OF SOLUTIONS

- When A is 2×2 , algebraic calculations can be supplemented by a geometric description of a system's evolution.
- We can view the equation $x_{k+1} = Ax_k$ as a description of what happens to an initial point x_0 in $\mathbb{R}^2$ as it is transformed repeatedly by the mapping $x \mapsto Ax$
- The graph of $x_0, x_1, \dots$ is called a **trajectory** of the dynamical system.

GRAPHICAL DESCRIPTION OF SOLUTIONS

- **Example 2** Plot several trajectories of the dynamical system $x_{k+1} = Ax_k$, when

$$A = \begin{bmatrix} .80 & 0 \\ 0 & .64 \end{bmatrix}$$

- **Solution** The eigenvalues of A are .8 and .64, with eigenvectors $v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. If $x_0 = c_1 v_1 + c_2 v_2$, then

$$x_k = c_1 (.8)^k \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c_2 (.64)^k \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

- Of course, x_k tends to 0 because $(.8)^k$ and $(.64)^k$ both approach 0 as $k \rightarrow \infty$. But the way x_k goes toward 0 is interesting. See Fig. 1 on the next slide.

GRAPHICAL DESCRIPTION OF SOLUTIONS

- Figure 1 shows the first few terms of several trajectories that begin at points on the boundary of the box with corners at $(\pm 3, \pm 3)$. The points on each trajectory are connected by a thin curve, to make the trajectory easier to see.

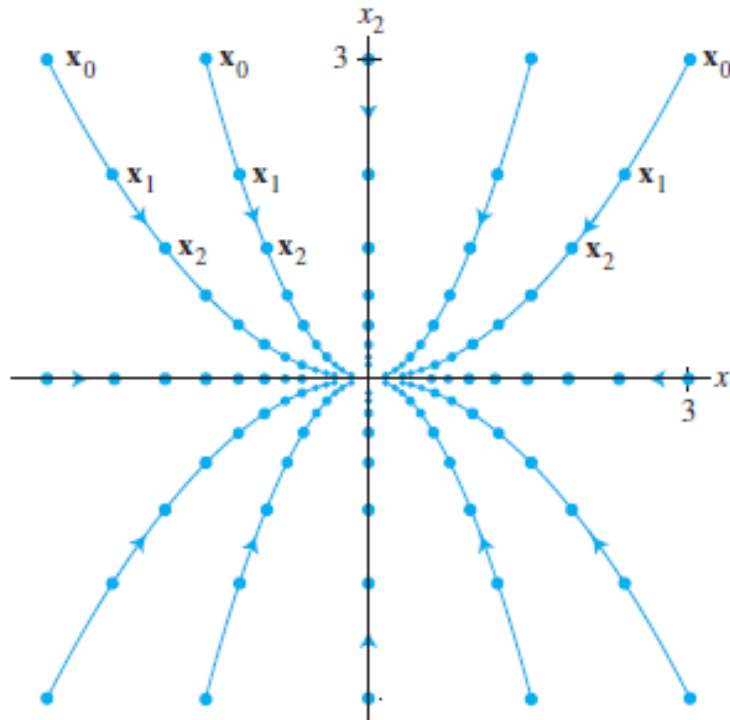


FIGURE 1 The origin as an attractor.

GRAPHICAL DESCRIPTION OF SOLUTIONS

- In example 2, the origin is called an **attractor** of the dynamical system because all trajectories tend toward 0.
- In the next example, both eigenvalues of A are larger than 1 in magnitude, and 0 is called a **repeller** of the dynamical system.
- **Example 3** Plot several typical solutions of the equation $\mathbf{x}_{k+1} = A\mathbf{x}_k$, where

$$A = \begin{bmatrix} 1.44 & 0 \\ 0 & 1.2 \end{bmatrix}$$

GRAPHICAL DESCRIPTION OF SOLUTIONS

- **Solution** The eigenvalues of A are 1.44 and 1.2. If $\mathbf{x}_0 = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$, then

$$\mathbf{x}_k = c_1(1.44)^k \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c_2(1.2)^k \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

- Both terms grow in size, but the first term grows faster.
- So the direction of greatest repulsion is the line through 0 and the eigenvector for the eigenvalue of larger magnitude.
- Fig. 2 on the next slide shows several trajectories that begin at points quite close to 0.

GRAPHICAL DESCRIPTION OF SOLUTIONS

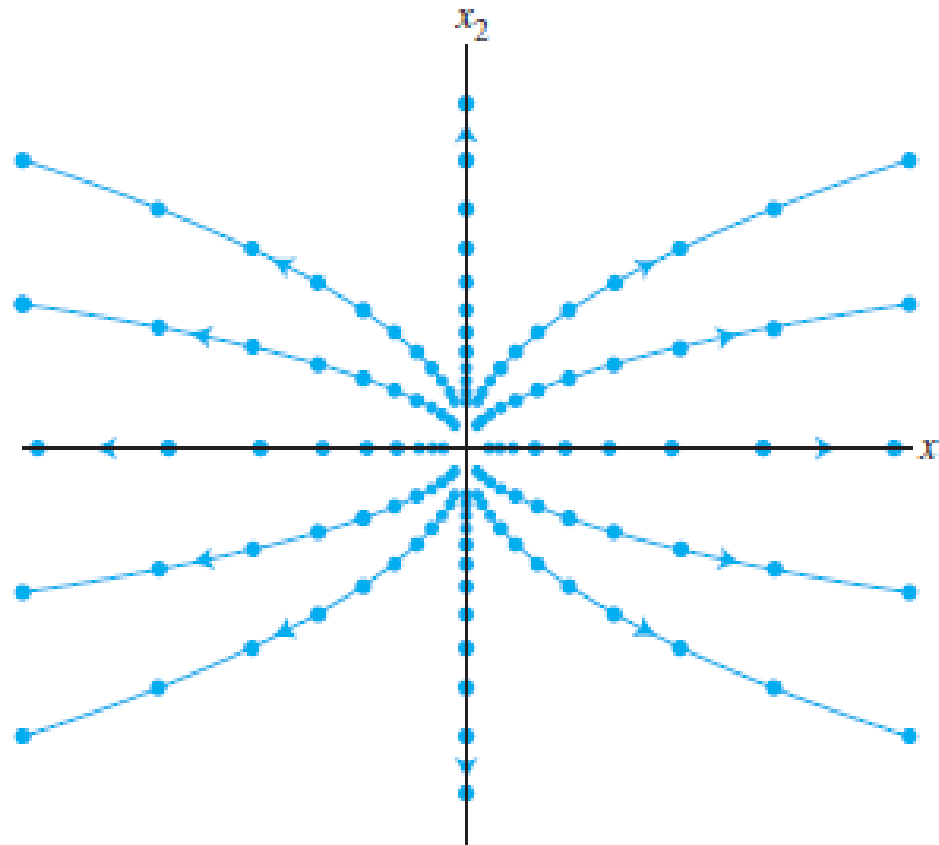


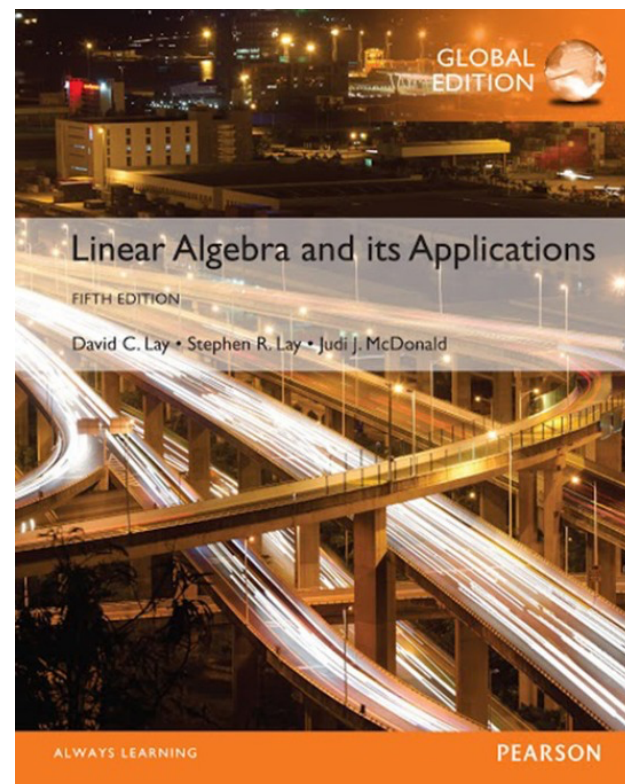
FIGURE 2 The origin as a repeller.

5

Eigenvalues and Eigenvectors

5.8

ITERATIVE ESTIMATES FOR EIGENVALUES



THE POWER METHOD

- The power method applies to an $n \times n$ matrix A with a **strictly dominant eigenvalue** λ_1 , which means that λ_1 must be larger in absolute value than all the other eigenvalues.
- **Example 1** Let $A = \begin{bmatrix} 1.8 & .8 \\ .2 & 1.2 \end{bmatrix}$, $v_1 = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$, and $x = \begin{bmatrix} -.5 \\ 1 \end{bmatrix}$. Then A has eigenvalues 2 and 1, and the eigenspace for $\lambda_1 = 2$ is the line through 0 and v_1 .
- For $k = 0, \dots, 8$, compute $A^k x$ and construct the line through 0 and $A^k x$. What happens as k increases?

THE POWER METHOD

- **Solution** The first three calculations are

$$Ax = \begin{bmatrix} 1.8 & .8 \\ .2 & 1.2 \end{bmatrix} \begin{bmatrix} -.5 \\ 1 \end{bmatrix} = \begin{bmatrix} -.1 \\ 1.1 \end{bmatrix}$$

$$A^2x = \begin{bmatrix} 1.8 & .8 \\ .2 & 1.2 \end{bmatrix} \begin{bmatrix} -.1 \\ 1.1 \end{bmatrix} = \begin{bmatrix} .7 \\ 1.3 \end{bmatrix}$$

$$A^3x = \begin{bmatrix} 1.8 & .8 \\ .2 & 1.2 \end{bmatrix} \begin{bmatrix} .7 \\ 1.3 \end{bmatrix} = \begin{bmatrix} 2.3 \\ 1.7 \end{bmatrix}$$

- Analogous calculations complete Table 1.

TABLE 1 Iterates of a Vector

k	0	1	2	3	4	5	6	7	8
$A^k \mathbf{x}$	$\begin{bmatrix} -.5 \\ 1 \end{bmatrix}$	$\begin{bmatrix} -.1 \\ 1.1 \end{bmatrix}$	$\begin{bmatrix} .7 \\ 1.3 \end{bmatrix}$	$\begin{bmatrix} 2.3 \\ 1.7 \end{bmatrix}$	$\begin{bmatrix} 5.5 \\ 2.5 \end{bmatrix}$	$\begin{bmatrix} 11.9 \\ 4.1 \end{bmatrix}$	$\begin{bmatrix} 24.7 \\ 7.3 \end{bmatrix}$	$\begin{bmatrix} 50.3 \\ 13.7 \end{bmatrix}$	$\begin{bmatrix} 101.5 \\ 26.5 \end{bmatrix}$

THE POWER METHOD

- The vectors $\mathbf{x}, A\mathbf{x}, \dots, A^4\mathbf{x}$ are shown in Fig. 1 below:

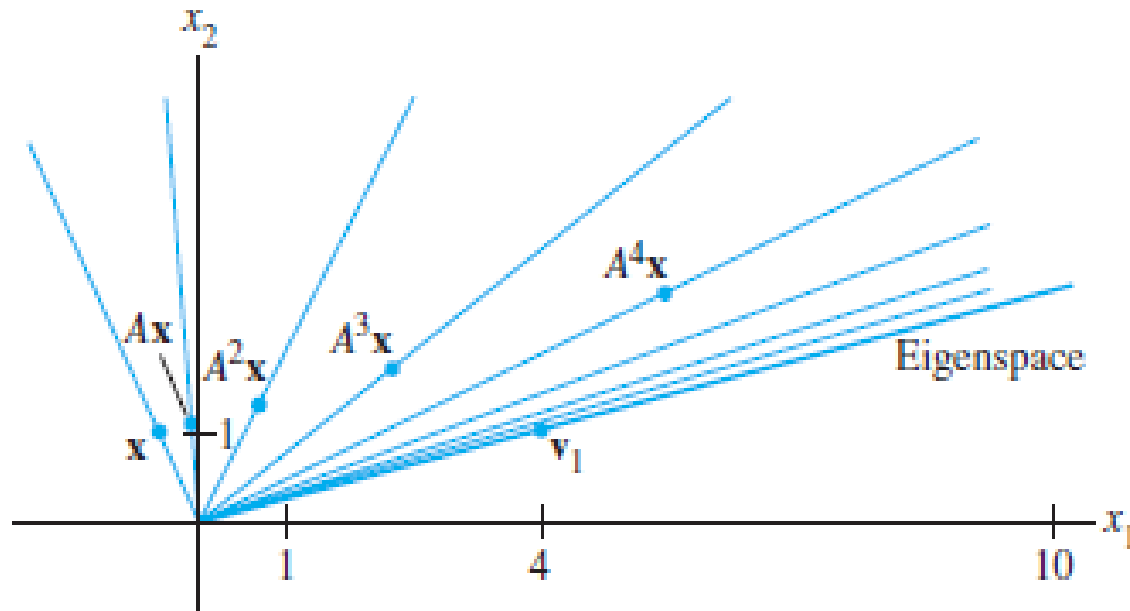


FIGURE 1 Directions determined by $\mathbf{x}, A\mathbf{x}, A^2\mathbf{x}, \dots, A^7\mathbf{x}$.

- The angle between the line (subspace) determined by $A^k\mathbf{x}$ and the line (eigenspace) determined by $\mathbf{v}_1$ goes to zero as $k \mapsto \infty$.

THE POWER METHOD

- **The power method for estimating a strictly dominant eigenvalue**
 1. Select an initial vector x_0 , whose largest entry is 1.
 2. For $k = 0, 1, \dots$,
 - a) Compute Ax_k .
 - b) Let μ_k be an entry in Ax_k whose absolute value is as large as possible.
 - c) Compute $x_{k+1} = (1/\mu_k)Ax_k$.
 3. For almost all choices of x_0 , the sequence $\{\mu_k\}$ approaches the dominant eigenvalue, and the sequence $\{x_k\}$ approaches a corresponding eigenvector.

THE INVERSE POWER METHOD

- **The inverse power method for estimating an eigenvalue λ of A**
 1. Select an initial estimate α sufficiently close to λ .
 2. Select an initial vector $\mathbf{x}_0$ whose largest entry is 1.
 3. For $k = 0, 1, \dots$,
 - a) Solve $(A - \alpha I)\mathbf{y}_k = \mathbf{x}_k$ for $\mathbf{y}_k$.
 - b) Let μ_k be an entry in $\mathbf{y}_k$ whose absolute value is as large as possible.
 - c) Compute $v_k = \alpha + (1/\mu_k)$
 - d) Compute $\mathbf{x}_{k+1} = (1/\mu_k)\mathbf{y}_k$.
 4. For almost all choices of $\mathbf{x}_0$, the sequence $\{v_k\}$ approaches the eigenvalue λ of A , and the sequence $\{\mathbf{x}_k\}$ approaches a corresponding eigenvector.

THE INVERSE POWER METHOD

- **Example 3** It is not uncommon in some applications to need to know the smallest eigenvalue of a matrix A and to have at hand rough estimates of the eigenvalues.
- Suppose 21, 3.3, and 1.9 are estimates for the eigenvalues of the matrix A below. Find the smallest eigenvalue, accurate to size decimal places.

$$A = \begin{bmatrix} 10 & -8 & -4 \\ -8 & 13 & 4 \\ -4 & 5 & 4 \end{bmatrix}$$

THE INVERSE POWER METHOD

- **Solution** The smallest eigenvalues seem close together so we use the inverse power method for $A - 1.9I$. Results of a MATLAB calculation are shown in Table 3 below:

TABLE 3 The Inverse Power Method

k	0	1	2	3	4
$\mathbf{x}_k$	$\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$	$\begin{bmatrix} .5736 \\ .0646 \\ 1 \end{bmatrix}$	$\begin{bmatrix} .5054 \\ .0045 \\ 1 \end{bmatrix}$	$\begin{bmatrix} .5004 \\ .0003 \\ 1 \end{bmatrix}$	$\begin{bmatrix} .50003 \\ .00002 \\ 1 \end{bmatrix}$
$\mathbf{y}_k$	$\begin{bmatrix} 4.45 \\ .50 \\ 7.76 \end{bmatrix}$	$\begin{bmatrix} 5.0131 \\ .0442 \\ 9.9197 \end{bmatrix}$	$\begin{bmatrix} 5.0012 \\ .0031 \\ 9.9949 \end{bmatrix}$	$\begin{bmatrix} 5.0001 \\ .0002 \\ 9.9996 \end{bmatrix}$	$\begin{bmatrix} 5.000006 \\ .000015 \\ 9.999975 \end{bmatrix}$
μ_k	7.76	9.9197	9.9949	9.9996	9.999975
ν_k	2.03	2.0008	2.00005	2.000004	2.0000002

THE INVERSE POWER METHOD

- Here x_0 was chosen arbitrarily, $y = (A - 1.9I)^{-1}x_k$, μ_k is the largest entry in y_k , $v_k = 1.9 + 1/\mu_k$, and $x_{k+1} = (1/\mu_k)y_k$.
- As it turns out, the initial eigenvalue estimate was fairly good, and the inverse power sequence converged quickly.
- The smallest eigenvalue is exactly 2.